

SIMATS ENGINEERING
CO 4 ASSESSMENT TOOL 3 – CASE STUDY

COURSE CODE: ITA14

COURSE NAME: Ethical Hacking

COURSE OUTCOME COVERED

CO4: Implement network security testing methods by performing web server attacks, analyzing web application vulnerabilities, and assessing wireless network security using Kali Linux. (BT4, BT5)

Assessment Tool Weightage: 35%

QUESTIONS: 5

Total Marks:100

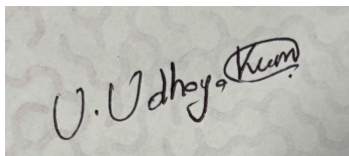
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1. A mid-sized e-commerce company hosts its web application on an Apache server running on a Linux environment. Recently, customers reported unauthorized transactions and abnormal account activities. Upon investigation, the IT team discovered suspicious HTTP requests in server logs, including attempts to access hidden directories and injection of malicious scripts into input fields. Additionally, the server is running an outdated version with known vulnerabilities, and no Web Application Firewall (WAF) is implemented. As a cybersecurity analyst, you are tasked with conducting a comprehensive security assessment using Kali Linux. Analyze the possible attack vectors exploited by the attacker, including web server vulnerabilities, input validation issues, and outdated software components. Explain in detail how you would perform reconnaissance, vulnerability scanning, and exploitation using appropriate Kali Linux tools. Further, propose a layered defense strategy that includes server hardening, secure coding practices, and network-level protections to prevent such incidents in the future.
 2. A financial services web application allows users to log in, view account details, and perform transactions. A recent penetration test revealed that the application is vulnerable to SQL injection and Cross-Site Scripting (XSS) attacks. Error messages expose database structure information, and user inputs are not properly sanitized. Attackers have exploited these weaknesses to extract sensitive customer data and inject malicious scripts into web pages viewed by other users. You are assigned to perform a detailed security audit of the application. Question: Examine how SQL injection and XSS vulnerabilities can be exploited in this scenario. Describe a step-by-step methodology using Kali Linux tools such as Burp Suite and SQLmap to identify and validate these vulnerabilities. Evaluate the impact of such attacks on confidentiality, integrity, and availability (CIA triad). Recommend comprehensive mitigation strategies, including input validation, output encoding, secure session handling, and implementation of security controls.
 3. A corporate office provides wireless internet access to employees using WPA2-PSK authentication. Recently, several employees reported unusual login prompts and slow network performance. A security investigation revealed the presence of a rogue access point (Evil Twin) mimicking the official network SSID. Attackers used this setup to capture login credentials and perform man-in-the-middle (MITM) attacks. You are tasked with analyzing the breach and strengthening wireless security. Explain how the attacker successfully executed the Evil Twin attack and intercepted sensitive data. Describe how Kali Linux tools such as Aircrack-ng and Wireshark can be used to detect and analyze such attacks. Evaluate the weaknesses in the existing wireless security configuration. Propose robust countermeasures, including secure authentication mechanisms, network monitoring, and user awareness strategies.

4. An organization's internal network hosts a web-based employee management system accessible over Wi-Fi. The wireless network uses weak passwords, and the web application lacks proper authentication and session management. An attacker gains access to the Wi-Fi network, performs reconnaissance, and exploits vulnerabilities in the web application to escalate privileges and access confidential employee data. You are assigned to perform a full penetration testing assessment. Analyze the attack chain starting from wireless network compromise to web application exploitation. Describe each phase of the attack, including reconnaissance, scanning, exploitation, and privilege escalation. Demonstrate how Kali Linux tools can be used in each phase. Propose a comprehensive security framework that addresses both wireless and web application vulnerabilities, ensuring defense-in-depth.
5. A university campus experiences frequent Wi-Fi disruptions during peak hours. Investigation reveals repeated deauthentication attacks targeting access points, causing legitimate users to disconnect. The attacker uses automated tools to continuously send deauth packets, leading to denial of service. You are tasked with identifying the cause and implementing solutions. Explain how deauthentication attacks are carried out in wireless networks and how attackers exploit protocol weaknesses. Describe how Kali Linux tools can be used to simulate and analyze such attacks in a controlled environment. Evaluate the impact on network availability and user productivity. Recommend effective mitigation strategies, including network configuration changes, intrusion detection systems, and secure wireless protocols.

Code of Conduct:

I UDHAYAKUMAR U (192424332)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A photograph of a handwritten signature in black ink on a light-colored, textured surface. The signature reads 'U. Udhay Kumar' in a cursive style, with 'Kumar' written in a slightly larger, more stylized font.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity difficult to understand

1. Security Assessment of an Apache-Based E-Commerce Web Application

Introduction

The e-commerce company is facing unauthorized transactions and abnormal account activity. The investigation has identified suspicious HTTP requests, attempts to access hidden directories, malicious scripts submitted through input fields, an outdated Apache server, and the absence of a Web Application Firewall (WAF).

The incident may involve multiple attack vectors operating together. A comprehensive security assessment should therefore examine the web server, application code, input validation, software versions, authentication mechanisms, and network protections.

Possible Attack Vectors

1. Outdated Web Server Vulnerabilities

The Apache server is running an outdated version. Older versions may contain publicly documented vulnerabilities that attackers can exploit to obtain unauthorized access, disclose information, bypass security controls, or cause denial of service.

2. Directory and File Enumeration

The suspicious requests attempting to access hidden directories indicate possible reconnaissance. Attackers may search for:

- Administrative directories

-

- Backup files

-

- Configuration files

-

- Temporary files

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- Old application versions

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- Exposed source-code files

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If sensitive files are accessible, attackers may obtain credentials or application information.

3. Input Validation Vulnerabilities

Malicious scripts inserted into input fields suggest a possible Cross-Site Scripting (XSS) vulnerability. If user input is stored or reflected without proper validation and output encoding, an attacker may cause malicious JavaScript to execute in another user's browser.

4. Authentication and Session Attacks

Unauthorized transactions may also indicate weaknesses in:

- Password policies
- Session management
- Multi-factor authentication
- Authorization checks
- Transaction validation
- CSRF protection
-

5. Vulnerable Software Components

Third-party libraries, plugins, frameworks, and operating-system packages may contain known vulnerabilities. An attacker can exploit the weakest component rather than the Apache server itself.

Reconnaissance Methodology Using Kali Linux

Reconnaissance should first be performed within the approved assessment scope.

Step 1: Identify the Target

Determine the authorized:

-

IP addresses

-

Domain names

-

Web applications

-

Network ranges

-

Testing time window

Step 2: Information Gathering

Kali Linux tools such as **Whois**, **DNS utilities**, **Nmap**, and **WhatWeb** can help identify publicly exposed services and technologies.

Nmap can be used to determine:

- Open ports
- Running services
- Service versions
- Operating-system information
-

WhatWeb can help identify web technologies such as:

- Apache
- PHP
- Web frameworks
- JavaScript libraries
-

Step 3: Web Enumeration

Tools such as **Gobuster** or **Dirsearch** can be used in an authorized lab to identify exposed directories and files.

The objective is to determine whether sensitive paths are publicly accessible, not to damage or modify the application.

Vulnerability Scanning

Nmap NSE, **Nikto**, and vulnerability scanners can be used to identify:

- Outdated software
- Dangerous HTTP configurations
- Missing security headers
- Known server weaknesses
- Exposed services
-

The discovered Apache version should be compared against official security advisories and the organization's patch-management records.

Controlled Exploitation

Exploitation should only be performed against a test environment or explicitly authorized production target.

For example, if a scanner identifies an outdated Apache component, the analyst can validate the vulnerability using a controlled proof-of-concept that demonstrates the security impact without extracting customer information or disrupting the service.

For web vulnerabilities, **Burp Suite** can intercept requests and help test whether input parameters are properly validated.

Layered Defense Strategy

Server Hardening

-

Upgrade Apache and Linux packages.

-

Remove unnecessary services.

-

Disable directory listing.

-

Remove unused modules.

-

Apply secure file permissions.

-

Protect configuration files.

-

Enable HTTPS.

-

Configure security headers.

-

Implement centralized logging.

-

Secure Coding

Developers should:

-

Validate input using allowlists.

-

Encode output according to its context.

-

Use parameterized database queries.

-

Implement CSRF protection.

-

Apply secure session management.

-

Enforce authorization checks.

-

Validate financial transactions on the server.

-

Web Application Firewall

A WAF can inspect HTTP traffic and block suspicious requests associated with:

- SQL injection
- XSS
- Path traversal
- Malicious request patterns
- Automated attacks
-

Network-Level Protection

A defense-in-depth architecture should include:

Internet → Firewall → WAF → Load Balancer → Web Server → Application Server → Database

Additional controls should include:

- Network segmentation
- IDS/IPS
- Secure administrative access
- Rate limiting
- Continuous monitoring

-

Centralized SIEM

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Regular vulnerability assessments

Conclusion

The incident is likely the result of multiple weaknesses rather than a single vulnerability. Updating the Apache server, securing application inputs, improving authentication and authorization, deploying a WAF, and implementing network monitoring can significantly reduce the attack surface.

2. SQL Injection and XSS Assessment of a Financial Web Application

Introduction

The financial application contains SQL injection and Cross-Site Scripting vulnerabilities. Error messages also expose database information, making reconnaissance easier for attackers. Because the application processes financial and customer information, these weaknesses can seriously affect confidentiality, integrity, and availability.

SQL Injection

SQL injection occurs when untrusted user input is incorporated into a database query without appropriate parameterization.

For example, an insecure application may conceptually construct a query like:

```
SELECT account FROM users WHERE username = '<user_input>';
```

If the input is directly inserted into the query, specially crafted input may change the intended SQL logic.

Possible Impact

An attacker may potentially:

- Bypass authentication
- Read unauthorized records
- Modify application data
- Delete database information
- Discover database structure
- Access sensitive customer information
-

The exact impact depends on database permissions and application design.

XSS

Cross-Site Scripting occurs when untrusted input is returned to a user's browser without appropriate output encoding.

For example, an attacker may place malicious content into:

-

Profile fields

-

Comments

-

Search fields

-

Transaction descriptions

-

Support forms

-

If the application stores that input and later displays it to other users, it can become a stored XSS vulnerability.

Potential XSS Impact

XSS may allow an attacker to:

-

Execute unauthorized scripts in a victim's browser

-

Manipulate page content

-

Perform actions using the victim's session

-

Attempt session-related attacks

-

Redirect users to malicious pages

-

Steal sensitive information accessible through the browser

-

Step-by-Step Security Audit Using Kali Linux

Step 1: Define Scope

Identify:

- Authorized URLs
- Test accounts
- Application functions
- API endpoints
- Database-backed input fields
-

Step 2: Burp Suite Proxy Configuration

Burp Suite can be used as an intercepting proxy.

The tester can:

Configure the browser to use Burp.

- Browse the application normally.
- Capture requests.
- Identify parameters.
- Send suspicious parameters to Burp Repeater.
- Compare normal and modified responses.

Step 3: Identify SQL Injection Points

Potential injection points include:

- Login fields
- Search parameters
- Product IDs
- Account IDs
- URL parameters
- API parameters
-

The tester should observe whether unusual input causes:

- Database errors
- Unexpected responses
- Different result sets
- Authentication anomalies
-

Testing should use non-destructive validation.

Step 4: SQLmap Validation

SQLmap can be used in an authorized testing environment to validate suspected SQL injection points.

A safe methodology is:

- Capture a test request using Burp Suite.
- Save the request for analysis.
- Provide the request to SQLmap.
- Allow SQLmap to determine whether an injectable parameter exists.
- Record the vulnerability evidence.
- Avoid destructive options and avoid accessing unnecessary customer records.
-

SQLmap should be treated as a validation tool rather than a justification for extracting sensitive production data.

Step 5: XSS Testing

Burp Suite can be used to identify parameters that accept HTML or script-like input.

The tester determines whether the application:

- Rejects unsafe input
- Encodes output
- Sanitizes content
- Stores the content
- Executes it when displayed
-

Testing should use harmless proof-of-concept markers in an authorized environment.

CIA Triad Analysis

Security Property Impact

Confidentiality	Customer and financial information may be exposed
Integrity	Transactions or account information may potentially be modified
Availability	Database manipulation or resource-intensive attacks may disrupt services

Mitigation Strategies

Input Validation

Use:

-

Allowlists

-

Data-type validation

-

Length restrictions

-

Range validation

-

Server-side validation

-

Parameterized Queries

Applications should use prepared statements rather than dynamically constructing SQL queries.

Output Encoding

HTML, JavaScript, URL, and other contexts require appropriate context-specific encoding.

Secure Session Management

Implement:

-

Secure cookies

-

HttpOnly cookies

-

SameSite settings

-

Session expiration

-

Session rotation

-

Reauthentication for sensitive operations

-

Multi-factor authentication

-

Error Handling

Database errors should never expose:

- SQL statements
- Table names
- Database versions
- File paths
- Internal stack traces
-

Users should receive generic error messages while detailed information is logged securely for administrators.

Additional Controls

- WAF
- IDS/IPS
- Secure development lifecycle
- SAST/DAST
- Regular penetration testing
- Database least privilege

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Security logging

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SIEM monitoring

Conclusion

SQL injection and XSS represent serious application-layer vulnerabilities. Burp Suite can be used to inspect and validate HTTP behavior, while SQLmap can help confirm SQL injection in an authorized environment. Strong input validation, parameterized queries, output encoding, secure sessions, and layered security controls are essential.

3. Evil Twin Attack Against a WPA2-PSK Wireless Network

Introduction

An Evil Twin attack occurs when an attacker creates a fraudulent wireless access point that imitates a legitimate network. In this scenario, employees see a familiar SSID and may connect to the rogue AP. The attacker can then attempt to capture authentication information or intercept traffic.

How the Attack Works

Step 1: Identify the Legitimate Network

The attacker observes wireless information such as:

- SSID
- Access-point identity
- Channel
- Signal characteristics
- Authentication configuration

Step 2: Create a Rogue Access Point

The attacker creates a wireless network using a name similar to or identical to the legitimate SSID.

The victim may see two networks with the same or similar name.

Step 3: Victim Connects

A user may accidentally connect to the stronger or more attractive rogue network.

This is particularly dangerous when employees automatically connect to known SSIDs without verifying the network.

Step 4: Traffic Interception

Once connected to the rogue network, the attacker may attempt to position themselves between the user and legitimate services.

Unencrypted traffic is particularly vulnerable. HTTPS and properly configured certificate validation provide important protection against interception.

Why WPA2-PSK Alone Is Not Enough

WPA2-PSK protects wireless communication when correctly configured, but organizational security can still be weakened by:

-

Shared passwords

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Weak PSKs

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Password reuse

-

Lack of individual user authentication

-

Poor rogue-AP detection

-

Automatic client association

-

Lack of network segmentation

Aircrack-ng for Wireless Analysis

The **Aircrack-ng** suite can be used in an authorized lab to analyze wireless networks.

It can help security teams:

-

Monitor wireless networks

-

Identify access points

-

Analyze wireless frames

-

Identify duplicate SSIDs

-

Examine authentication behavior

-

Validate wireless security configuration

-

The goal of the assessment should be identifying rogue infrastructure and weaknesses rather than capturing real employee credentials

Wireshark Analysis

Wireshark can analyze captured wireless traffic and help identify:

-

Management frames

-

Authentication events

-

Association events

- Repeated connection attempts
- Unexpected access points
- Unusual traffic patterns
- Unencrypted application traffic

A security analyst can compare the legitimate AP's characteristics against suspicious APs.

Security Weaknesses

The existing configuration is vulnerable because:

- WPA2-PSK uses a shared credential.
- Employees may not distinguish legitimate and rogue APs.
- There is inadequate wireless monitoring.
- There is no effective rogue-AP detection.
- MITM-resistant authentication may not be implemented.
- Network segmentation may be insufficient.

Countermeasures

Use WPA2/WPA3-Enterprise

Enterprise authentication provides individual credentials instead of one shared password.

A stronger architecture is:

Employee Device → WPA3/WPA2-Enterprise → RADIUS → Identity System

Certificate-Based Authentication

802.1X/EAP methods with proper certificate validation can help prevent users from unknowingly connecting to fraudulent networks.

Wireless Intrusion Detection

Deploy Wireless IDS/WIPS to detect:

- Rogue APs
- Evil Twins
- Unauthorized clients
- Suspicious wireless behavior
- Duplicate SSIDs
-

Network Segmentation

Separate:

- Employee devices
- Guest devices
- IoT devices
- Administrative systems
- Critical servers
-

User Awareness

Employees should be trained to:

-

Verify wireless networks.

-

Avoid suspicious login prompts.

-

Report duplicate SSIDs.

-

Avoid entering credentials into unexpected pages.

-

Report unusual Wi-Fi behavior.

Conclusion

The Evil Twin attack demonstrates that wireless security requires more than encryption. Enterprise authentication, certificate validation, wireless monitoring, network segmentation, endpoint security, and employee awareness should work together to reduce the risk.

4. Penetration Testing of a Wireless Network and Employee Web Application

Introduction

This scenario represents a multi-stage attack chain. The attacker first compromises weak Wi-Fi security and then uses network access to discover and exploit weaknesses in the employee management application.

The complete attack chain can be represented as:

Weak Wi-Fi → Network Access → Reconnaissance → Scanning → Web Enumeration → Vulnerability Identification → Exploitation → Privilege Escalation → Data Access

Phase 1: Reconnaissance

The penetration tester first identifies the authorized wireless and internal network scope.

Information collected may include:

- Wireless SSIDs
- Access points
- Internal IP ranges
- DNS information
- Available services
- Web applications
-

Kali Linux Tools

Nmap can identify authorized hosts and services.

Wireshark can analyze network traffic in the test environment.

Wireless analysis tools can identify nearby access points and their security configurations.

Phase 2: Wireless Security Assessment

The weak Wi-Fi password is assessed to determine whether it meets organizational security requirements.

The assessment should examine:

-

Password strength

-

Authentication method

-

Encryption

-

WPS configuration

-

Guest-network isolation

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Client isolation

-

Enterprise authentication availability

-

Aircrack-ng can be used in a controlled laboratory to analyze wireless security and validate configuration weaknesses.

Phase 3: Network Scanning

After obtaining authorized network access, Nmap can be used to identify:

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Active hosts

-

Open ports

-

HTTP/HTTPS services

-

SSH services

-

Database services

-

Other exposed applications

-

The result should be documented as an asset inventory.

Phase 4: Web Application Enumeration

The tester examines the employee management application.

Areas of interest include:

-

Login page

-

Employee profiles

-

Administrative pages

-

Search functions

-

APIs

-

File upload functionality

-

Session handling

-

Access-control mechanisms

-

Burp Suite can capture and analyze HTTP requests and responses.

Phase 5: Vulnerability Identification

The tester checks for:

- Weak authentication
- Broken access control
- IDOR
- SQL injection
- XSS
- CSRF
- Session weaknesses
- Insecure file upload
- Security misconfiguration

Tools such as Burp Suite, Nmap, Nikto, and appropriate vulnerability scanners can support the assessment.

Phase 6: Exploitation

Only non-destructive exploitation should be performed against the authorized application.

For example, if a normal employee can access an administrator-only endpoint, the tester can demonstrate the authorization failure using test accounts.

The objective is to prove the vulnerability without modifying real employee information.

Phase 7: Privilege Escalation

Privilege escalation occurs when a low-privileged account obtains higher privileges without authorization.

The tester should evaluate whether:

- Roles are enforced server-side.
- Administrative endpoints are protected.
- Object identifiers are properly authorized.
- Session roles can be modified.
- API permissions are correctly implemented.

For example:

Employee Account → Unauthorized Administrative Function → Elevated Access

would indicate a broken access-control vulnerability.

Phase 8: Impact Analysis

Possible consequences include:

- Unauthorized employee-data access
- Modification of employee records
- Exposure of personal information
- Administrative account compromise
- Further internal network attacks
-

Defense-in-Depth Framework

Wireless Layer

- WPA3-Enterprise
- 802.1X authentication
- Strong credentials
- Disable unnecessary WPS
- Wireless IDS/WIPS
-

Rogue AP detection

Network Layer

-

VLAN segmentation

-

Network access control

-

Firewall rules

-

IDS/IPS

-

Zero-trust principles

-

Least-privilege network access

-

Application Layer

-

Strong authentication

-

MFA

-

Secure session management

-

RBAC

-

Server-side authorization

Data Layer

-

Encryption

-

Database least privilege

-

Backup

-

Access monitoring

-

Data-loss prevention

Monitoring Layer

Implement:

Wireless Monitoring → Network IDS → Web Logs → Application Logs → SIEM → Security Alert

Conclusion

The attack demonstrates why wireless and application security cannot be treated independently. A compromised wireless network can provide the initial foothold, while a vulnerable web application can allow privilege escalation and sensitive-data access. Defense-in-depth must therefore protect every stage of the attack chain.

5. Deauthentication Attack and Wireless Denial of Service

Introduction

A deauthentication attack is a wireless denial-of-service technique in which an attacker causes wireless clients to disconnect from an access point. Repeated attacks can prevent legitimate users from maintaining stable wireless connections.

This is especially serious in a university environment because students, faculty, and staff depend on Wi-Fi for academic and administrative activities.

How Deauthentication Attacks Work

Traditional Wi-Fi management frames were not always strongly authenticated. An attacker can exploit weaknesses in the handling of management traffic by transmitting forged deauthentication frames.

The simplified process is:

Client Connected to AP → Forged Deauthentication Frame → Client Disconnects → Client Attempts Reconnection → Repeated Deauthentication → Service Disruption

An automated attacker can repeat this process continuously.

Protocol Weakness

The fundamental weakness is associated with the protection of wireless management frames.

If management frames are not adequately protected, an attacker may be able to generate frames that appear to originate from a legitimate access point.

Impact

Availability

The largest impact is on availability.

Users may experience:

-

Repeated disconnections

-

Slow network performance

-

Inability to access online resources

-

Failed authentication

-

Interrupted video classes

-

Interrupted file transfers

-

Loss of productivity

Operational Impact

University IT staff may also experience:

-

Increased support requests

-

Network troubleshooting workload

-

Difficulties identifying the attacker

-

Reduced service reliability

Kali Linux Controlled-Lab Analysis

The attack should be simulated only against equipment owned by the organization or in an isolated laboratory.

Step 1: Wireless Monitoring

Aircrack-ng tools can be used to observe:

-

Access points

-

Wireless clients

-

Channels

-

Management frames

-

Connection behavior

-

Step 2: Packet Capture

Wireshark can capture wireless traffic for analysis.

The analyst can look for:

-

Large numbers of deauthentication frames

-

Repeated disconnections

-

Unusual frame rates

-

Unexpected transmitter addresses

-

Correlation between deauthentication frames and client disconnections

Step 3: Controlled Simulation

In an isolated lab, authorized security personnel can generate representative management-frame traffic to determine whether monitoring systems detect it.

The objective should be:

Generate Test Event → Detect Event → Alert Security Team → Record Evidence → Apply Mitigation

rather than disrupting a production university network.

Detection Strategy

A Wireless Intrusion Detection System can monitor:

- Deauthentication-frame frequency
- Abnormal management traffic
- Rogue APs
- Unknown transmitters
- Repeated client disconnections
-

SIEM integration can provide centralized alerting.

For example:

Wireless Sensor → WIDS/WIPS → SIEM → Security Alert → Incident Response

Mitigation Strategies

1. Protected Management Frames

Deploy **802.11w / Protected Management Frames (PMF)** where supported.

WPA3 networks provide stronger support for protected management frames.

2. WPA3

Move from older wireless configurations toward WPA3 where compatible with the organization's devices.

3. Wireless IDS/WIPS

Deploy wireless security monitoring capable of detecting:

-

Deauthentication floods

-

Rogue APs

-

Evil Twins

-

Unauthorized wireless devices

-

Abnormal management traffic

4. Network Segmentation

Separate:

-

Students

-

Faculty

-

Administration

-

Guests

-

IoT devices

-

This limits the impact of a wireless security incident.

5. Access Point Configuration

Organizations should:

-

Use enterprise authentication.

-

Disable unnecessary wireless features.

-

Maintain updated firmware.

-

Use appropriate channel planning.

-

Monitor AP health.

6. Incident Response

When an attack is detected:

- Identify affected APs.
- Confirm abnormal management traffic.
- Identify the suspected transmitting device.
- Preserve logs and packet captures.
- Isolate unauthorized equipment according to organizational procedures.
- Restore stable wireless service.
- Review security configuration.
- Document the incident.

Conclusion

Deauthentication attacks primarily threaten wireless availability. The combination of protected management frames, WPA3/enterprise authentication, wireless intrusion detection, network segmentation, continuous monitoring, and incident response can significantly reduce the impact of these attacks.